



Poisson Distribution · Worksheet 1

Further Statistics 1 Chapter 2: Poisson Distribution. Answer all questions. Show your working clearly. Total: 69 marks. Time: 85 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are three sections, one for each topic in Chapter 2. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Give probabilities to 4 decimal places. State the distribution you use, with its parameter(s), e.g. $X \sim \text{Po}(3.5)$.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Poisson probabilities and conditions	20	
B Adding Poisson distributions and changing the interval	22	
C Poisson approximation to the binomial	22	
Roar-level challenge	5	
Total	69	

Section A Poisson probabilities and conditions

20 marks

1. Given that $X \sim \text{Po}(3)$, find $P(X = 2)$. [2]

.....

.....

2. Given that $X \sim \text{Po}(4.5)$, find (a) $P(X \leq 3)$, (b) $P(X > 5)$. [3]

.....

.....

.....



3. The number of emails arriving in Rex's inbox in one hour is to be modelled by a Poisson distribution. [3]
State three conditions that are needed for this model to be valid, in context.

.....

.....

.....

4. The random variable $X \sim \text{Po}(\lambda)$. Given that $P(X = 0) = 0.2$, [4]
(a) find the exact value of λ ,
(b) find $P(X = 2)$.

.....

.....

.....

.....

.....



5. Customers arrive at a shop at random, at an average rate of 6 every 10 minutes. The random variable X is the number of customers arriving in a 10-minute period.

(a) State two assumptions needed for X to be modelled by $Po(6)$. [2]

.....

.....

(b) Find $P(X = 4)$. [2]

.....

.....

(c) Find $P(X > 8)$. [2]

.....

.....

(d) Over 50 ten-minute periods, the numbers of arrivals have mean 6.1 and variance 11.2. [2]
Comment on the suitability of the Poisson model, giving a reason.

.....

.....

.....

(Total for Question 5 is 8 marks)



Section B Adding Poisson distributions and changing the interval

22 marks

6. Faults occur at random along a cable at a rate of 0.4 per metre. [2]
Find the probability that a 5-metre length of cable has no faults.

.....

.....

7. The random variables $X \sim \text{Po}(2)$ and $Y \sim \text{Po}(3)$ are independent. [3]
Find $P(X + Y = 4)$.

.....

.....

.....

8. Calls arrive at a switchboard at random, at a rate of 3 every 10 minutes. [3]
Find the probability that more than 20 calls arrive in one hour.

.....

.....

.....

9. A shop sells laptops at random at a mean rate of 1.5 per day, and, independently, tablets at a mean rate of 2.5 per day. [4]
- (a) Find the probability that, on one day, the shop sells a total of at least 3 laptops and tablets.
- (b) Find the probability that, over two days, the shop sells a total of exactly 5 laptops and tablets.

.....

.....

.....

.....

.....



10. A typist makes typing errors at random at a rate of 0.6 per page and, independently, spelling errors at random at a rate of 0.4 per page.

(a) Find the probability that a page has no errors of either type. [2]

.....

.....

(b) Find the probability that a chapter of 8 pages has more than 10 errors in total. [3]

.....

.....

.....

(c) Given that a page has exactly 2 errors, find the probability that both are typing errors. [3]

.....

.....

.....

(d) Five pages are chosen at random. Find the probability that at least 2 of them have no errors. [2]

.....

.....

.....

(Total for Question 10 is 10 marks)

**Section C Poisson approximation to the binomial**

22 marks

11. The random variable $X \sim B(n, p)$. State the conditions under which X may be approximated by a Poisson distribution, and give the parameter of that distribution. [2]

.....

.....

12. Given that $X \sim B(200, 0.015)$, use a Poisson approximation to find $P(X = 2)$. [3]

.....

.....

.....

13. Given that $X \sim B(150, 0.02)$, use a Poisson approximation to find $P(X > 4)$. [3]

.....

.....

.....

14. The random variable $X \sim B(60, 0.05)$. [4]

- (a) Find the exact value of $P(X \leq 2)$.
- (b) Use a Poisson approximation to find $P(X \leq 2)$.

.....

.....

.....

.....

.....



15. A machine makes bolts. Each bolt, independently, is defective with probability 0.015. The bolts are packed in boxes of 200.

- (a) Suggest a suitable approximate distribution for the number of defective bolts in a box, and justify your choice. [2]

.....

.....

- (b) Find the probability that a box contains exactly 4 defective bolts. [2]

.....

.....

- (c) A customer rejects a box if it contains more than 5 defective bolts. Find the probability that a box is rejected. [3]

.....

.....

.....

- (d) The customer buys 10 boxes. Find the probability that at most one of them is rejected. [3]

.....

.....

.....

(Total for Question 15 is 10 marks)



ROAR-LEVEL CHALLENGE

5 marks

16. The independent random variables X and Y are such that $X \sim \text{Po}(2)$ and $Y \sim \text{Po}(3)$. [5]

(a) Find $P(X = 2 \mid X + Y = 5)$.

(b) More generally, $X \sim \text{Po}(\lambda)$ and $Y \sim \text{Po}(\mu)$ are independent. Show that the distribution of X , given that $X + Y = n$, is $B\left(n, \frac{\lambda}{\lambda + \mu}\right)$.

.....

.....

.....

.....

.....

.....

.....

.....

.....

END OF WORKSHEET

Total: 69 marks